

EduCore360: role-based academic management

Comprehensive backend API and platform architecture guide

Project overview

System purpose

[cite_start]

EduCore360 is a robust REST API designed to handle the complex end-to-end workflows of academic institutions, ensuring secure and efficient management of educational data[cite: 5].

Key capabilities

[cite_start]

- Stateless authentication using JWT [cite: 6, 9]
[cite_start]
- Role-based access control (RBAC) [cite: 5, 9]
[cite_start]
- Automated class and exam scheduling [cite: 33, 34]
[cite_start]
- Academic performance tracking [cite: 34]

Technical specifications



Spring Boot 3

[cite_start]

Modern Java framework for rapid application development and simplified configuration[cite: 6].



JWT & Security

[cite_start]

Secured session management with digitally signed JSON Web Tokens for identity integrity[cite: 6, 9].



MySQL & JPA

[cite_start]

Relational data persistence with Hibernate ORM for seamless database-entity mapping[cite: 6, 8].

System architecture

[cite_start]

- ✓ **Presentation layer:** REST Controllers exposing role-specific endpoints with JSON serialization[cite: 11, 12, 13, 14].

[cite_start]

- ✓ **Service layer:** Dedicated business logic implementations enforcing academic constraints[cite: 11, 15, 16, 17].

[cite_start]

- ✓ **Repository layer:** Data access abstractions utilizing Spring Data JPA for indexed querying[cite: 11, 18, 19, 20].

[cite_start]

- ✓ **Database layer:** Persistent storage in MySQL with strict unique constraints and relations[cite: 11, 21, 22, 23].

User roles & access

[cite_start]

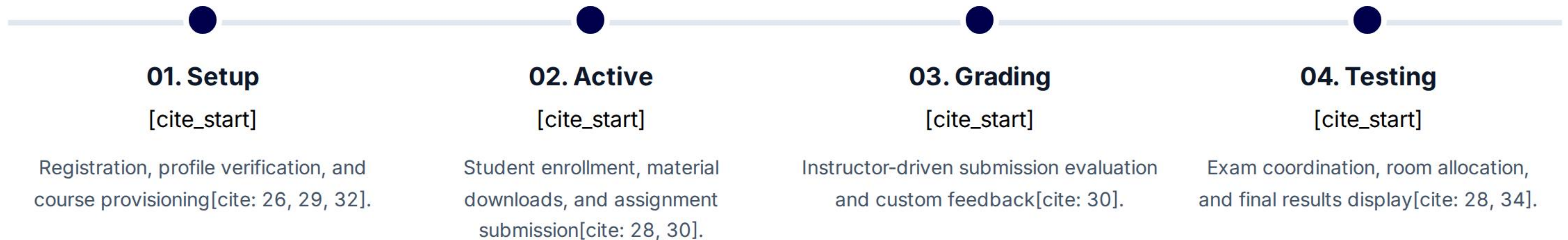
Defining responsibilities and security boundaries across the platform [cite: 24]

User role matrix

[cite_start] [cite_start] [cite_start] [cite_start]

Role	Core responsibility	Authorized actions
Student	Course participation	Enroll, download materials, submit tasks, take exams[cite: 27, 28].
Instructor	Curriculum delivery	Publish notes, grade tasks, manage exams[cite: 29, 30].
Registrar	Administration	Provision courses, assign staff, view enrollments[cite: 31, 32].
Exam Coord.	Testing management	Room allocation, schedule exams, publish results[cite: 33, 34].

End-to-end operational flow



Security & cross-cutting concerns

Identity protection

[cite_start]

Every request is intercepted by a custom Security Filter that extracts and validates the JWT claims before authorizing controller execution[cite: 6, 9].

Automated auditing

AOP-based logging captures critical system modifications, ensuring that every sensitive action is recorded in the audit log for accountability.

Stateless design philosophy

[cite_start]

"By offloading session state to the client via JWT, the application remains lightweight, secure, and ready for high-concurrency academic operations[cite: 6, 9]."

Questions?

Thank you for your attention.

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